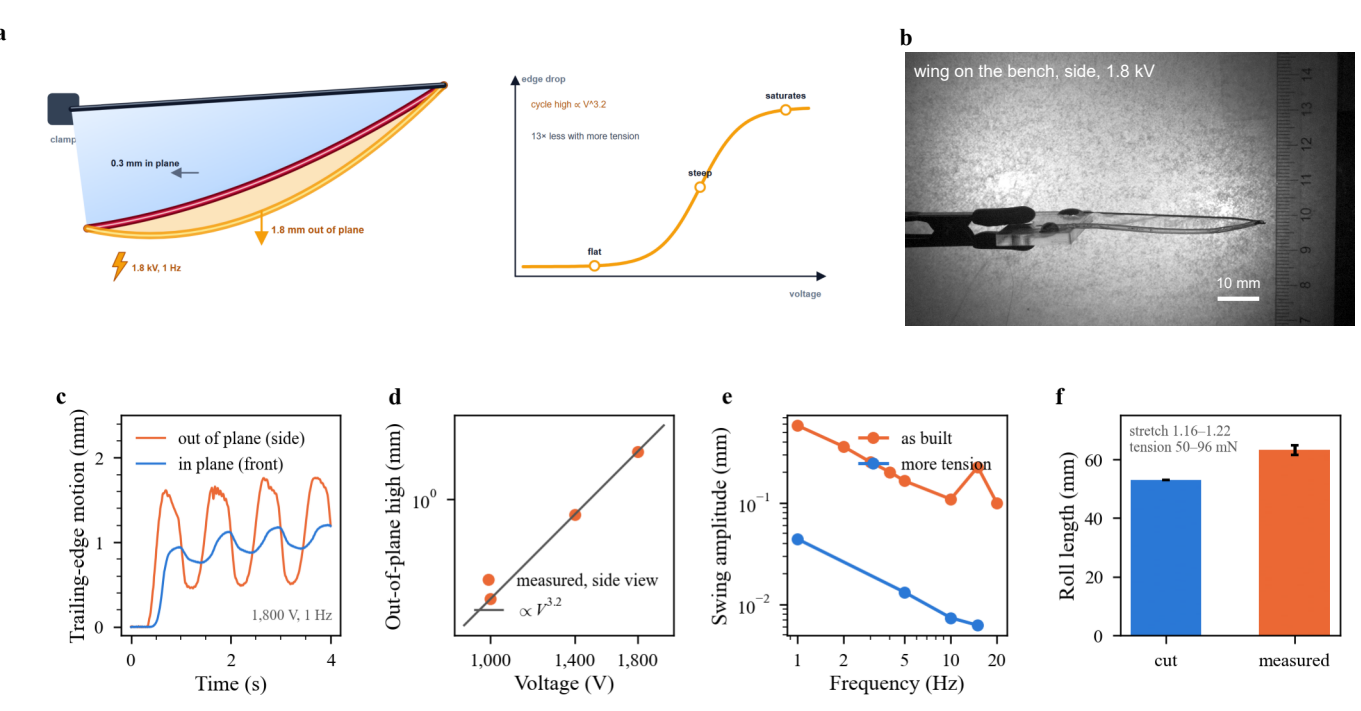

a
clamp
0.3 mm in plane
1.8 mm out of plane
1.8 kV, 1 Hz
edge drop
cycle high ∝ V^3.2
13× less with more tension
saturates
steep
flat
voltage
b
wing on the bench, side, 1.8 kV
10 mm
c
out of plane (side)
in plane (front)
1,800 V, 1 Hz
Trailing-edge motion (mm)
Time (s)
d
measured, side view
∝ V^3.2
Out-of-plane high (mm)
Voltage (V)
e
as built
more tension
Swing amplitude (mm)
Frequency (Hz)
f
stretch 1.16–1.22
tension 50–96 mN
Roll length (mm)
cut
measured